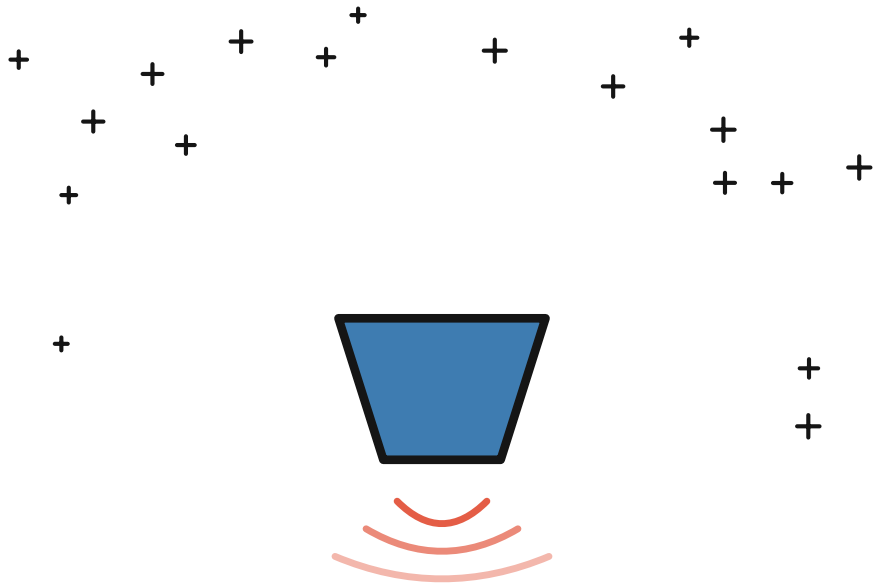


Next:

Coming Home

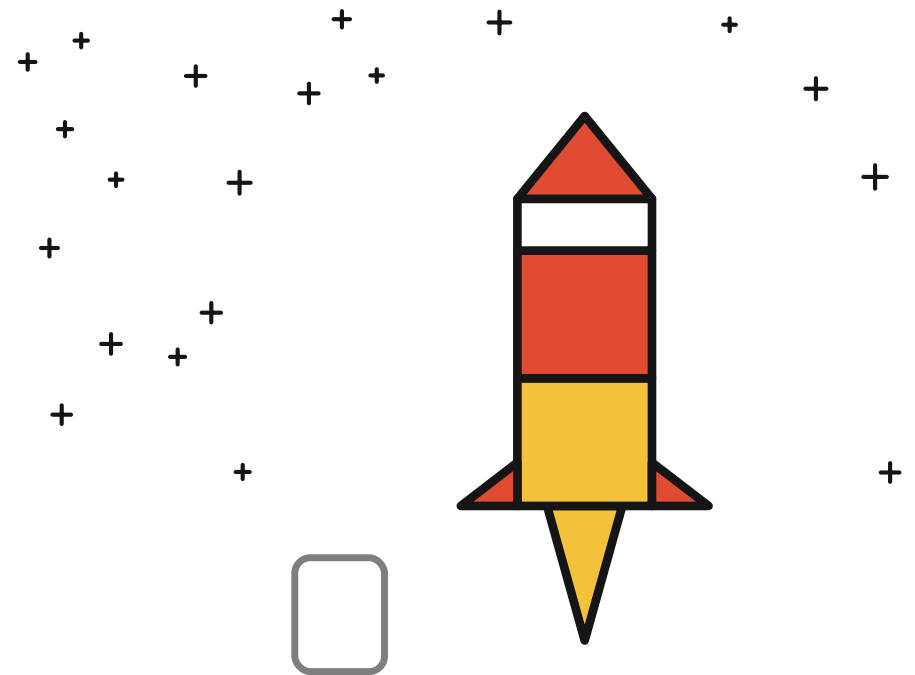
Book Six · why a blunt shape is safer than a sharp one



Going up is hard.
Stopping is harder.

TOO HEAVY TO LIFT

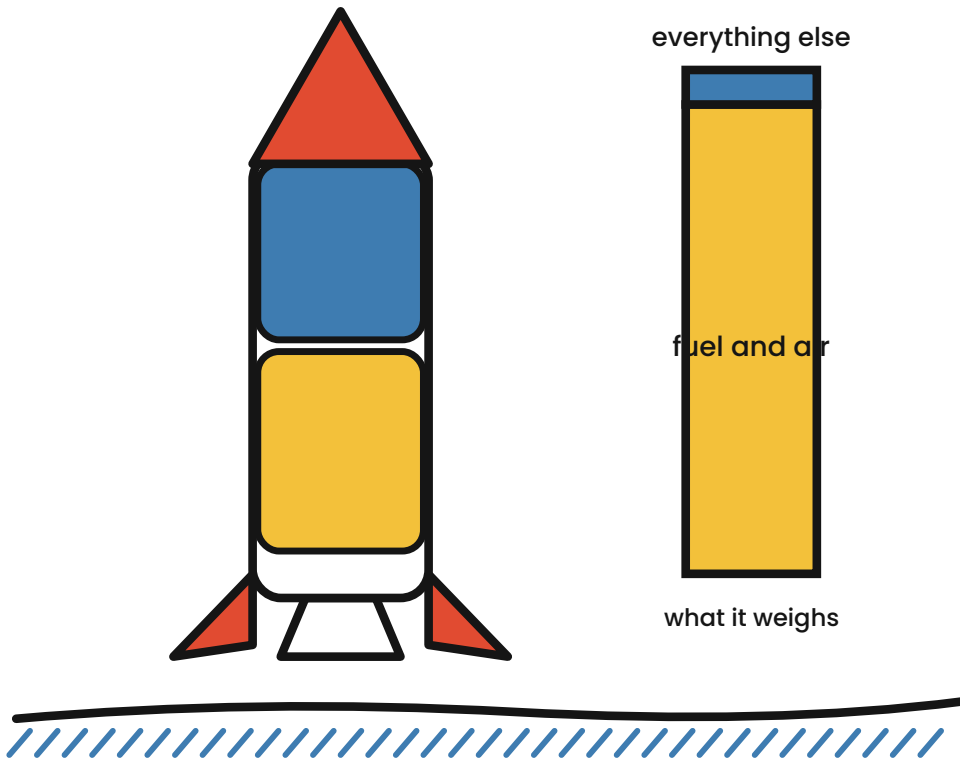
Book Five · Why a rocket throws itself away



falling away

Print it. Read it together. Then empty your pockets.

A rocket is mostly just fuel.



Nearly all of it is the stuff it burns.

For the grown-up.

The one idea

The rocket equation says the speed you gain is the exhaust velocity times the natural log of your mass ratio. Because it is a logarithm, wanting a little more speed costs enormously more rocket. Staging is the one honest way to cheat it: stop carrying tanks that are no longer doing anything.

The arithmetic on page seven

Same propellant, same payload, same engines. One stage gives about 4.8 km/s; three stages give about 6.3. That gain is bought purely by discarding dead structure. Tsiolkovsky proposed exactly this in 1903 and called them rocket trains.

Why even three stages fall short

Six point three is still under the 7.8 needed for orbit, and that gap is real. Actual launchers close it with structural efficiency rather than more stages — a modern first stage is around 95 percent propellant by mass, thinner-walled relative to its size than a drinks can. Staging was necessary but never sufficient.

Saturn V

110.6 metres, 2,970 tonnes at lift-off, three stages. The first dropped at about 67 km after 2 minutes 41 seconds; the second at roughly 185 km. The command module that came home weighed about 5.5 tonnes — around two parts in a thousand of what left the pad.

Try this tonight.

1 Carry the shopping upstairs.

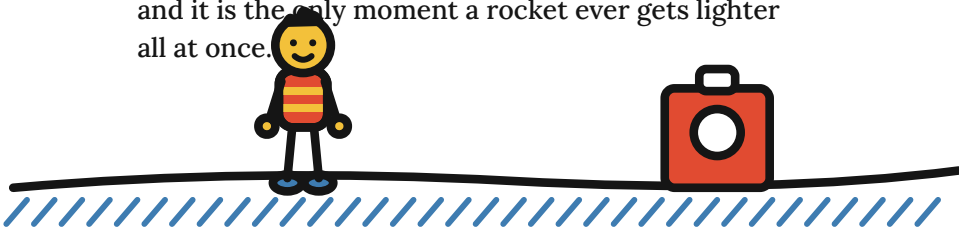
Take the bags up in one trip, then again in three, putting each empty bag down as you go. The second way is easier for exactly the reason a rocket sheds its stages.

2 Fill a bottle and lift it.

Lift an empty plastic bottle, then the same bottle full of water. Ask him which part got heavier. The bottle did not change at all — and that bottle is what a rocket refuses to keep carrying.

3 Stack three boxes and take one away.

Build a tower, then remove the bottom box and let the top two settle down. That is stage separation, and it is the only moment a rocket ever gets lighter all at once.



And fuel is heavy.

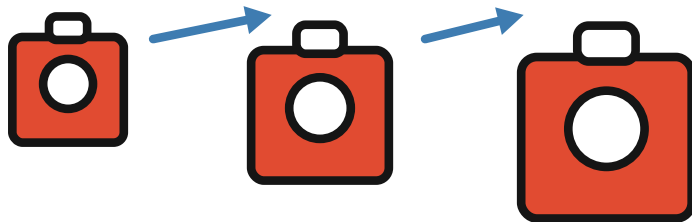
Try lifting a full can.



Now imagine a tank the size of a house.

So you need fuel to lift the fuel.

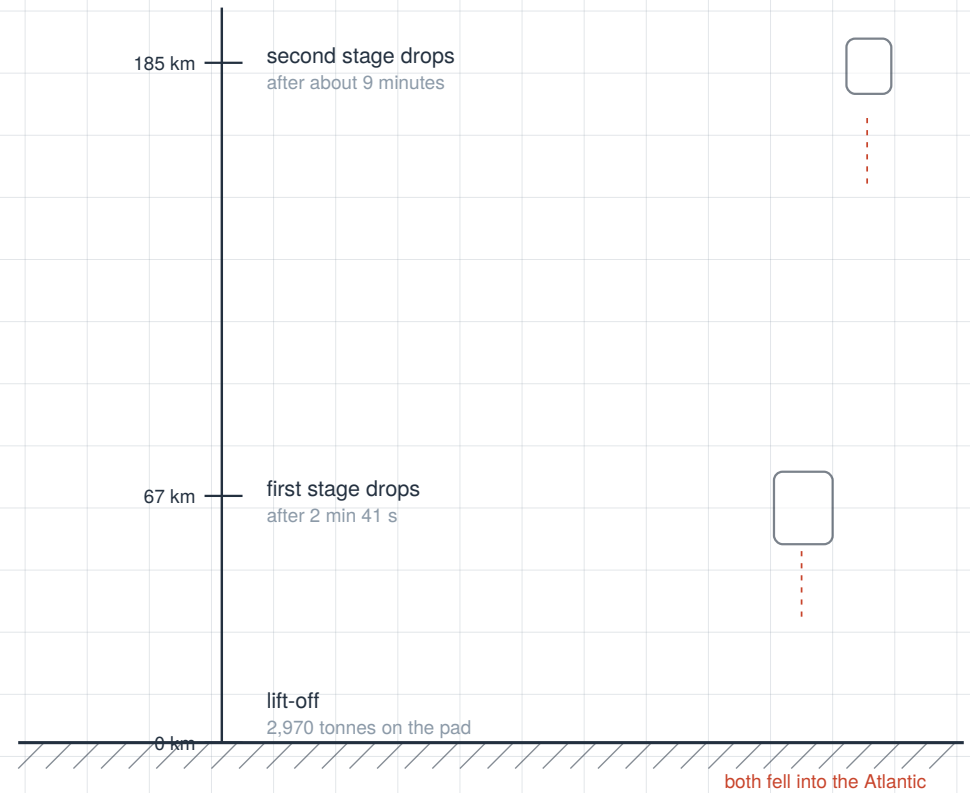
more fuel
means more weight
which needs more fuel



It never seems to stop.

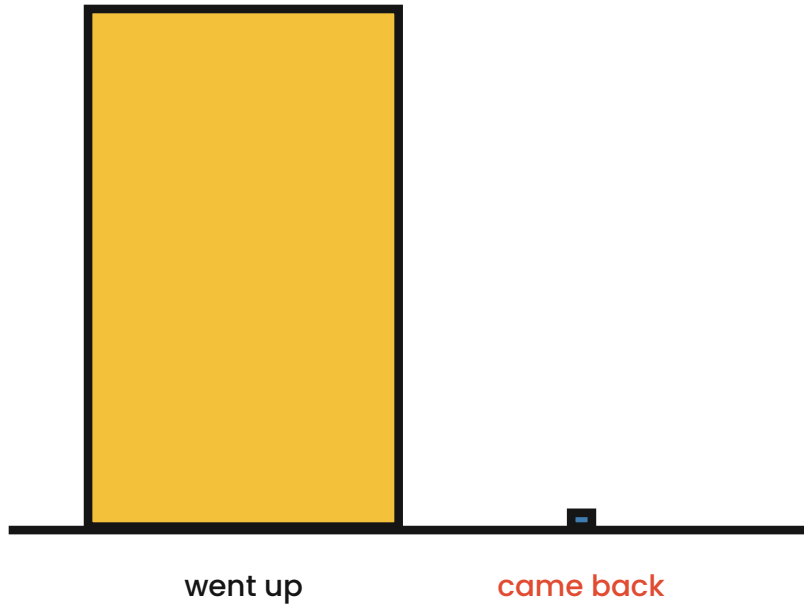
What fell away, and when.

Too Heavy to Lift - figure 2 - Saturn V, 1969



Of the 2,970 tonnes that left the pad, about 5.5 tonnes returned — roughly two parts in a thousand. Everything else was structure, propellant, or deliberately discarded. The third stage pushed on to the Moon.

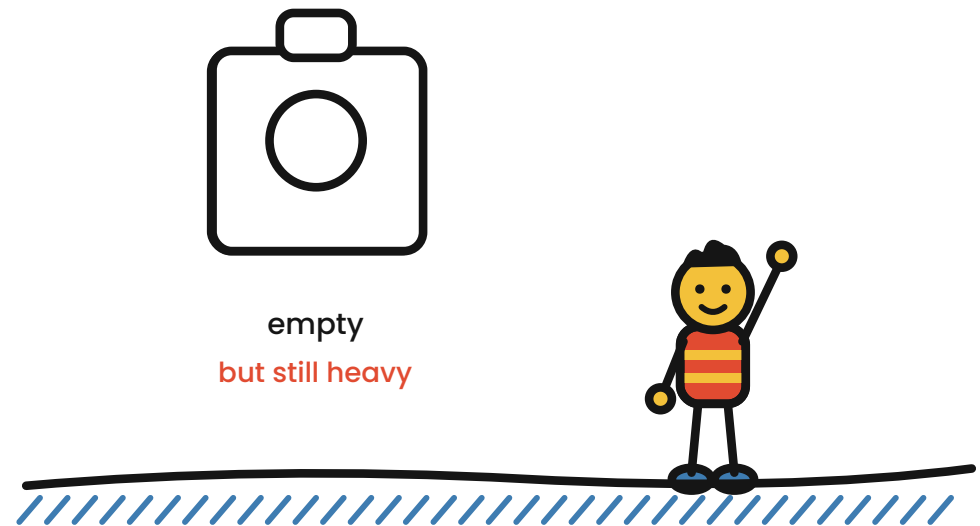
Three thousand tonnes
went up.



Five and a half tonnes came home.
That little square is all of it.

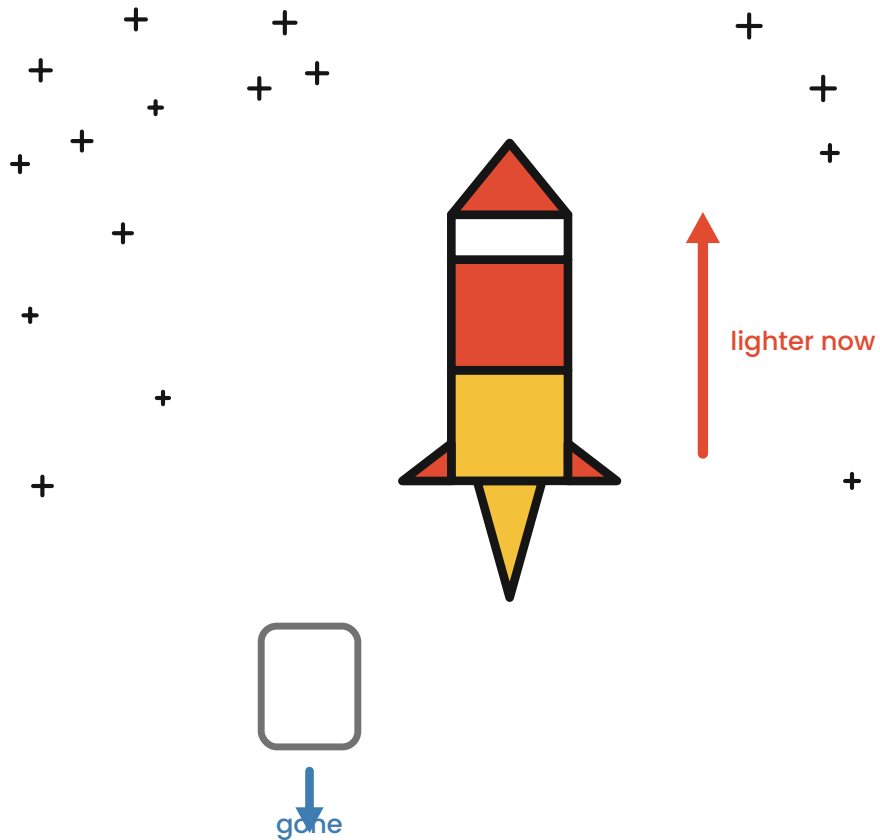
Here is the trick.

When a tank is empty, it is still there.



And you are still carrying it all the way up.

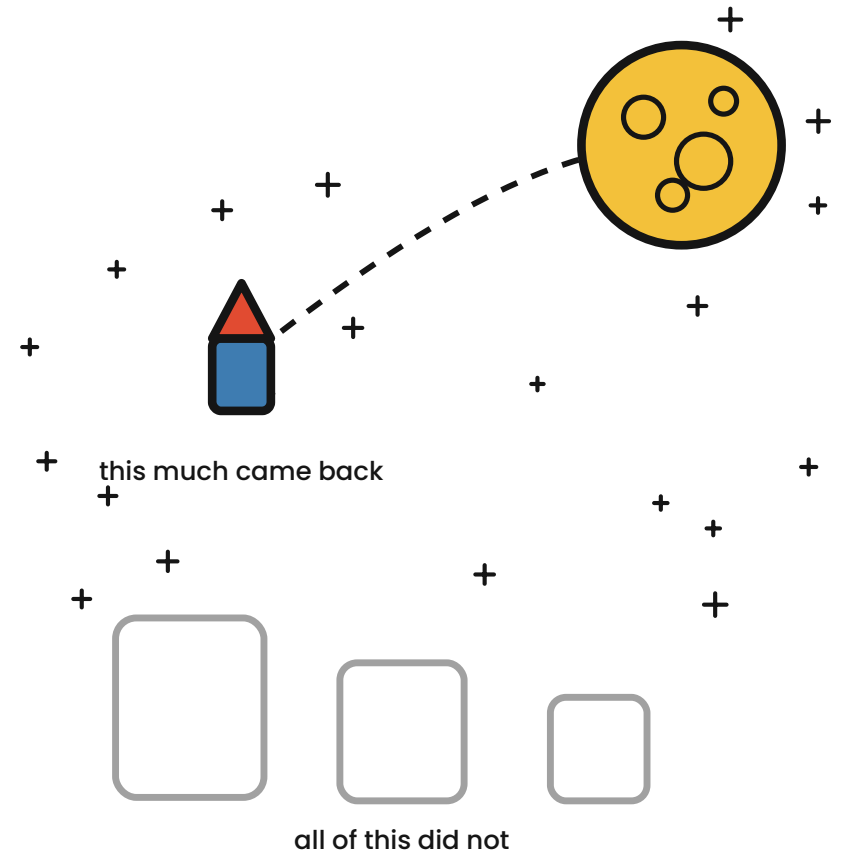
So let it fall away.



The moment a tank is empty, drop it.

It went to the Moon.

And it threw nearly all of itself away.



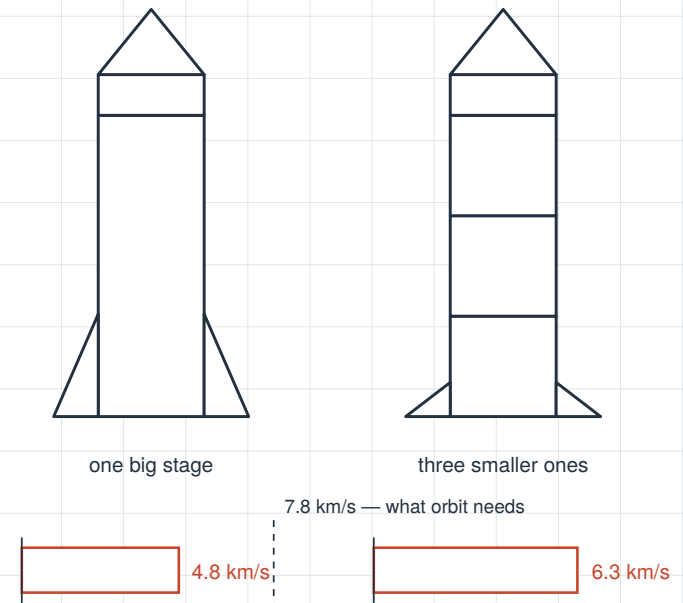
The biggest one ever was called Saturn V.



As tall as a building with thirty-six floors.

One piece, or three?

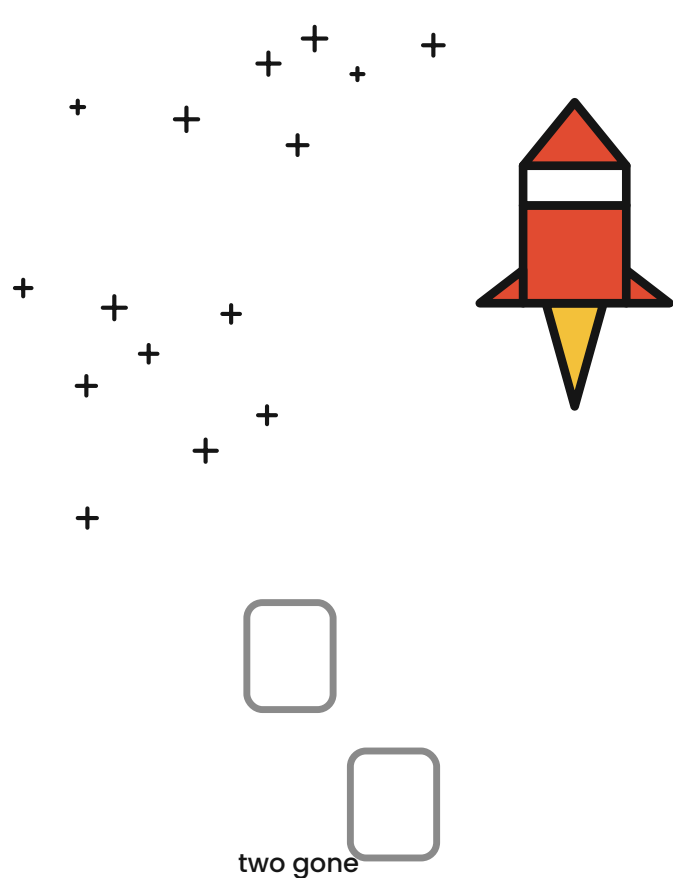
Too Heavy to Lift - figure 1 - the same fuel; split up



Identical propellant. Splitting it into three stages buys 1.5 km/s for nothing but the decision to stop carrying empty tanks.

Note that even three stages fall short. Staging alone was never enough.

Then do it again.



Lighter. Then lighter again.

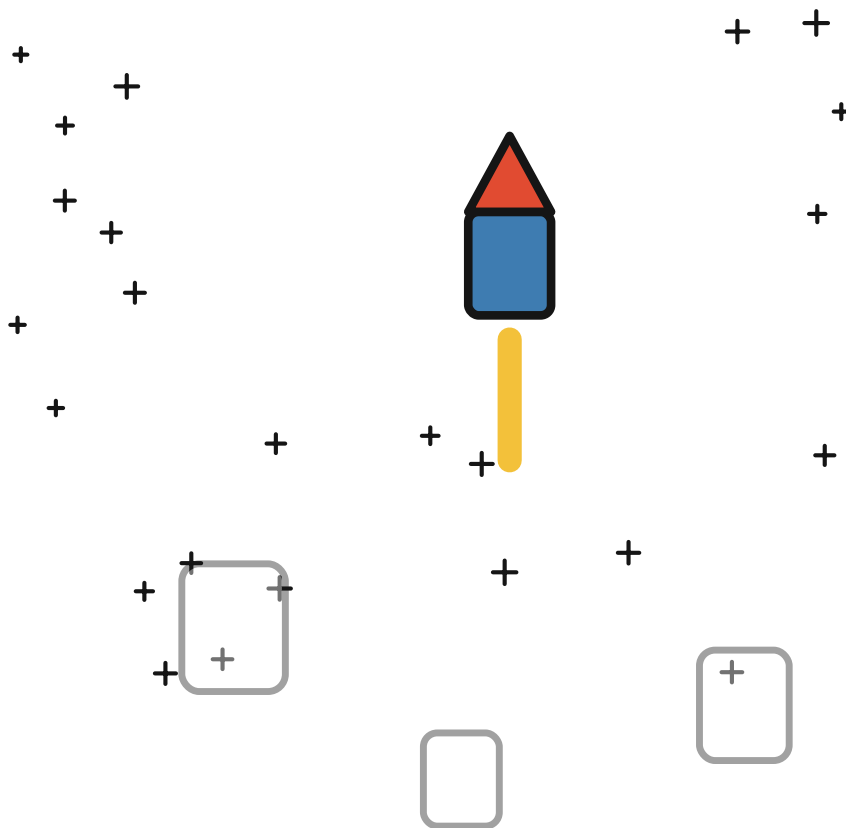
8

Only the little top
keeps going.

pierce here

pierce here

pierce here



Everything else is somewhere in the sea.

9

fold along this line